

The empirical recurrence for OEIS A198664, proved by a two-line transfer matrix

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Abstract

OEIS A198664 counts arrays in which every cell is constrained by its neighbours on *both* sides, and which are moreover written in canonical form (“values introduced in row major order”). Neither feature is visible to a one-line transfer matrix: the cell condition reaches the row above and the row below, and the canonical-form clause is not local at all. Both are dealt with. What is counted is equality patterns, so $5!a(n) = \sum_i \binom{5}{i} D_{5-i} L_i(n)$ with D_m the derangement numbers and L_i the count over an i -letter alphabet; and each L_i is a walk count whose vertices are ordered pairs of consecutive rows. Relabelling-invariance makes the chain strongly lumpable, cutting 20515 pairs down to $S = 544$ states. The entry’s empirical recurrence of order 60, contributed by R. H. Hardin, then follows from a finite exact computation.

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1 The sequence and the conjecture

OEIS A198664 is “Number of $n \times 3$ 0..4 arrays with values 0..4 introduced in row major order and each element equal to one or two horizontal and vertical neighbors.”. It has offset 1 and begins

1, 13, 132, 2715, 59281, 1544783, 40546053, 1100262051, ...

The entry carries the following comment:

Empirical: $a(n) = 26*a(n-1) + 207*a(n-2) - 4693*a(n-3) - 5776*a(n-4) + 157591*a(n-5) + 139443*a(n-6) - 1825139*a(n-7) - 3912555*a(n-8) + 9205948*a(n-9) + 49540346*a(n-10) - 50523258*a(n-11) - 187400427*a(n-12) + 193241548*a(n-13) + 457708054*a(n-14) - 2348260717*a(n-15) - 853094525*a(n-16) + 19413615836*a(n-17) - 4559327195*a(n-18) - 81422778316*a(n-19) + 29945338304*a(n-20) + 301610165665*a(n-21) - 167143208735*a(n-22) - 816553663024*a(n-23) + 863486594864*a(n-24) + 360655337798*a(n-25) - 4549562921429*a(n-26) + 2936432568750*a(n-27) + 19962871586477*a(n-28) - 6710978972167*a(n-29) - 45546401098564*a(n-30) + 12779999633307*a(n-31) + 41720387693559*a(n-32) - 36287192385505*a(n-33) + 1527013277477*a(n-34) + 152317108173902*a(n-35) + 35931632692232*a(n-36) - 236911335057032*a(n-37) - 134246505542044*a(n-38) - 202262924273384*a(n-39) - 284614527001184*a(n-40) + 24380741738928*a(n-41) + 465170070212768*a(n-42) + 610398531322432*a(n-43) + 205669857696256*a(n-44) + 965878890074624*a(n-45) - 10768073967104*a(n-46) + 541791030492160*a(n-47) + 1703742317537280*a(n-48) - 974079293448192*a(n-49) - 101946236010496*a(n-50) - 762059395989504*a(n-51) - 2488636072656896*a(n-52) - 986411874910208*a(n-53) - 226783416811520*a(n-54) - 358318577025024*a(n-55) + 403639919181824*a(n-56) + 362849171931136*a(n-57) + 64276333068288*a(n-58) - 22574348107776*a(n-59) - 14843406974976*a(n-60) for $n > 61$$

As of the “Last modified” line on the live entry (Oct 03 2025 07:59:05, revision 9) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) = 26a(n-1) + 207a(n-2) - 4693a(n-3) - 5776a(n-4) + 157591a(n-5) + 139443a(n-6) - 1825139a(n-7) \quad (1)$$

for all $n > 61$.

2 The condition, and what the canonical-form clause counts

The arrays have 3 columns and $L = n$ rows. Write $x_{t,u}$ for the entry in row t and column u , with the neighbour offsets

$$\mathcal{N} = \{(-1, 0), (1, 0), (0, -1), (0, 1)\}$$

written as (row shift, column shift), the requirement is

$$\#\{(d_1, d_2) \in \mathcal{N} : (t + d_1, u + d_2) \text{ lies in the array and its entry is equal to } x_{t,u}\} \in \{1, 2\} \quad (2)$$

for every cell (t, u) .

The condition is stated purely in equalities between entries, so it is unchanged by any relabelling of the values: it is a property of the *equality pattern* — the partition of the cells into classes of equal entries — and not of the values themselves. The clause “values introduced in row major order” selects exactly one array from each relabelling class, so with $K = 5$ available values the entry counts admissible patterns with at most K classes.

Lemma 1. *Let $L_i(n)$ count the arrays over an alphabet of i letters that satisfy the cell condition, with no canonical-form clause, and $N_j(n)$ the admissible patterns with exactly j classes. Then $L_i = \sum_j N_j i(i-1) \cdots (i-j+1)$ and*

$$5! \sum_{j \leq 5} N_j(n) = \sum_{i=0}^5 \binom{5}{i} D_{5-i} L_i(n), \quad D_m = m! \sum_{t=0}^m \frac{(-1)^t}{t!}.$$

Proof. A labelled array satisfying the condition is a pattern together with an injection of its classes into the alphabet, and a pattern with j classes admits $i(i-1) \cdots (i-j+1)$ of them; that is the first identity. Writing $i(i-1) \cdots (i-j+1) = j! \binom{i}{j}$ it reads $L_i = \sum_j (j! N_j) \binom{i}{j}$, so binomial inversion gives $j! N_j = \sum_i (-1)^{j-i} \binom{j}{i} L_i$. Summing over $j \leq 5$ and exchanging the order of summation, the coefficient of L_i is $\sum_{t \leq 5-i} (-1)^t / (i! t!)$, which multiplied by $5!$ is $\binom{5}{i} (5-i)! \sum_{t \leq 5-i} (-1)^t / t! = \binom{5}{i} D_{5-i}$. $\square$

3 Each L_i is a walk count on pairs of rows

Fix i and let $V_i = \{0, \dots, i-1\}^3$. Because (2) at a cell of row t involves rows $t-1$, t and $t+1$, a single row is not enough state. Take the ordered pairs $(p, c) \in V_i \times V_i$ as vertices and declare $(p, c) \rightarrow (c, x)$ an edge when every cell of the middle row c satisfies (2) with p above it and x below it. Let P_i be the adjacency matrix, $\mathbf{s}_i$ the indicator of the pairs (p, c) whose first row p satisfies the condition with no row above, and $\mathbf{e}_i$ the indicator of the pairs whose second row c satisfies it with no row below.

Lemma 2. *For $L \geq 2$, the arrays of L rows satisfying the condition are exactly the walks $(r^{(1)}, r^{(2)}) \rightarrow (r^{(2)}, r^{(3)}) \rightarrow \dots \rightarrow (r^{(L-1)}, r^{(L)})$ of length $L-2$ that start at a vertex marked by $\mathbf{s}_i$ and end at one marked by $\mathbf{e}_i$, so $L_i = \mathbf{s}_i^\top P_i^{L-2} \mathbf{e}_i$.*

Proof. A walk of that form is exactly a sequence of rows $r^{(1)}, \dots, r^{(L)}$, since consecutive vertices overlap in one row. In such a walk the condition at the cells of row t is tested once, at the step leaving $(r^{(t-1)}, r^{(t)})$, and with the correct neighbours $r^{(t-1)}$ and $r^{(t+1)}$; the cells of row 1 are tested by $\mathbf{s}_i$, which uses the condition with the row above absent, and those of row L by $\mathbf{e}_i$. So every cell is tested exactly once and with its true neighbourhood, which is the claim. $\square$

Lemma 3. *Write $\pi(p, c)$ for the equality pattern of the pair. If $\pi(p, c) = \pi(p', c')$ then for every pattern Q the two vertices have the same number of successors of pattern Q , and $\mathbf{s}_i, \mathbf{e}_i$ take the same value at both. Hence the chain is strongly lumpable over π and L_i is computed on the quotient, whose states are the set partitions of the $2 \cdot 3$ cells of a pair into at most i classes.*

Proof. Equal patterns means $(p', c') = (\sigma(p), \sigma(c))$ for a permutation σ of the alphabet. The edge condition and the two boundary conditions are statements about equalities between entries, so they are invariant under σ , and $x \mapsto \sigma(x)$ is a bijection between the successor sets preserving patterns. The number of pairs with a given pattern P is $e_P = i(i-1) \cdots (i-|P|+1)$, the number of injections of its classes into the alphabet, which supplies the weights on the quotient. $\square$

Assembling the 5 lumped chains block-diagonally into N of size $S = 544$, with the weights of Lemma 1 folded into the start vector $\mathbf{v}$ and the end-of-array indicator into $\mathbf{u}$,

$$5! a(n) = \mathbf{v}^\top N^{L-2} \mathbf{u}, \quad L = n. \quad (3)$$

4 The criterion, and the computation

Let $q(t) = t^{60} - \sum_i c_i t^{60-i}$ be the characteristic polynomial of (1) and put $G = N^1$, so that consecutive values of n advance the walk by 1 rows.

Lemma 4. *With $h_j = G^j N^0 \mathbf{u}$ for the offset o that aligns the first admissible index,*

$$5! \left(a(n) - \sum_i c_i a(n-i) \right) = \mathbf{v}^\top G^{j-60} \left(h_{60} - \sum_i c_i h_{60-i} \right)$$

for the corresponding j . Hence (1) holds from that point on if and only if $\mathbf{v}^\top G^m w = 0$ for every $m \geq 0$, where $w = h_{60} - \sum_i c_i h_{60-i}$; and it is enough to check $m = 0, 1, \dots, S-1$.

Proof. Substitute (3) and factor. The scalar $5!$ is nonzero, so it does not affect vanishing. For the last clause, Cayley–Hamilton applied to G makes G^S an integer combination of $I, G, \dots, G^{S-1}$, so each $\mathbf{v}^\top G^m w$ with $m \geq S$ is the same combination of the earlier values; if those vanish, so do all the rest. $\square$

Theorem 5. *The recurrence (1) holds for every $n > 61$, which is the statement of Section 1.*

Proof. The lumped transition counts were built by taking one representative pair for each pattern, enumerating every row x over the alphabet, testing (2) on the middle row, and sorting the results by the pattern of the new pair. The vector w was formed by 60 applications of G in exact integer arithmetic and $\mathbf{v}^\top G^m w$ evaluated for $m = 0, 1, \dots$, again exactly. Every one of those integers vanishes from the index corresponding to $n = 61 + 1$ onwards, and the run was continued until $S + 1$ consecutive zeros had been seen, which by Lemma 4 settles every larger m . $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the whole construction was compared against the entry's DATA: the right-hand side of (3), divided by $5!$, was evaluated at every one of the 16 published indices and agrees exactly, the quotient coming out an integer each time. This tests the modelling and not only the arithmetic — the inversion of Lemma 1, the reading of (2), the treatment of the first and last rows, and the alignment of the entry's index with $L = n$ would each show up as a mismatch.

Second, the lumped chain was checked against the unlumped one on the small shapes of this family: the walk counts over the full alphabet agree with $\sum_P e_P f_m(P)$ term by term.

Third, the recurrence (1) was evaluated on the published terms in exact integer arithmetic with no matrices involved, and the annihilation test of Lemma 4 was carried to the full Cayley–Hamilton bound rather than to a sampled prefix.

References

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